

Plant Root Phenotyping Pipeline.

ChronoRootAnalysis

Plant Analysis

Analysis Overview

Plant Overlay

Generate Report

Report

Individual plant root analysis

Select Project Folder

Select Video Folder

Raspberry Module

(the raspberry module identifier)

Camera

(the plate/camera identifier)

Plant Number

(a number to identify plants in the plate, left to right)

Identifier

(experiment identifier, e.g. WT, Col0)

Save

Preview video

Analyze Plant

Process all plants

Load config json from file

Load previous configuration

Analysis and postprocessing parameters

☐ Save Cropped Images

Manual Calibration Parameters:

Known (mm):

10

☐ Video has QR codes

Open Calibration Helper

Pixels:

240

Set processing limit

0

(in days, 0 means no limit)

Capture interval

15

(in minutes, usually 15 minutes)

Emergence distance

2

(in millimeters, default: 2 mm)

To begin the plant analysis process, launch the interface. This will open the main analysis window. Put your mouse over any button to see a description of what it does.

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(the raspberry module identifier)

(the plate/camera identifier)

(a number to identify plants in the plate, left to right)

Save

Preview video

Visual Inspection:

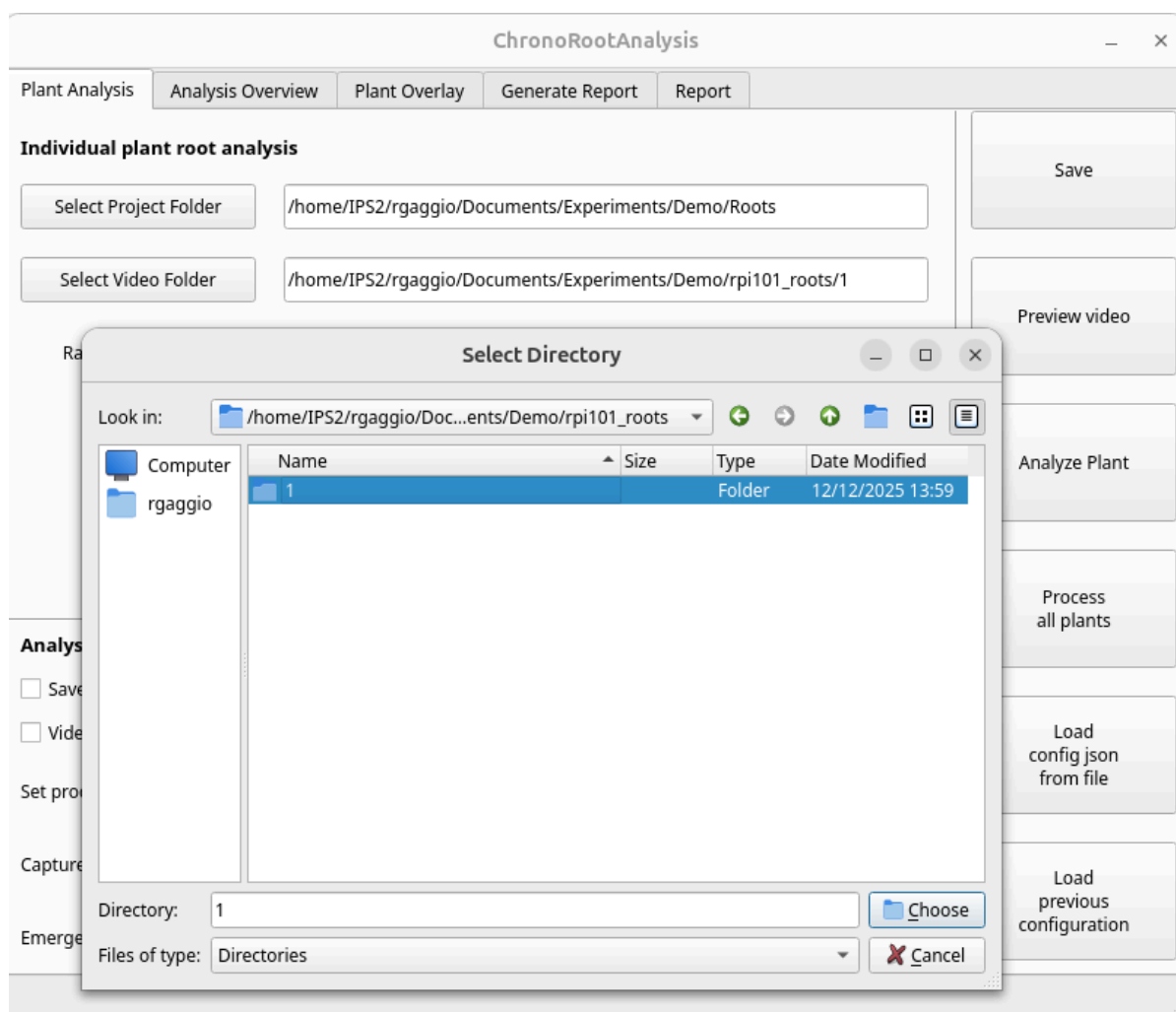
Open a viewer to check image quality and verify that the root segmentation is accurate.

Interface Components:

- **Select Project Folder:** Specifies the storage location for your analysis results. A project encompasses a complete experimental setup involving one or more Raspberry Pi modules.
- **Select Video Folder:** Identifies the video file for current processing. Videos must be processed sequentially, one at a time.
- **Plant Identification Fields:** The Raspberry Pi Module, Camera, Plant Number, and Identifier fields allow you to tag each plant according to its origin (camera and module), position on the plate, and biological variety.
- **Manual Calibration Parameters:** If your video does not have a 1-cm QR code, measure a known distance and write the pixels from the calibration helper.

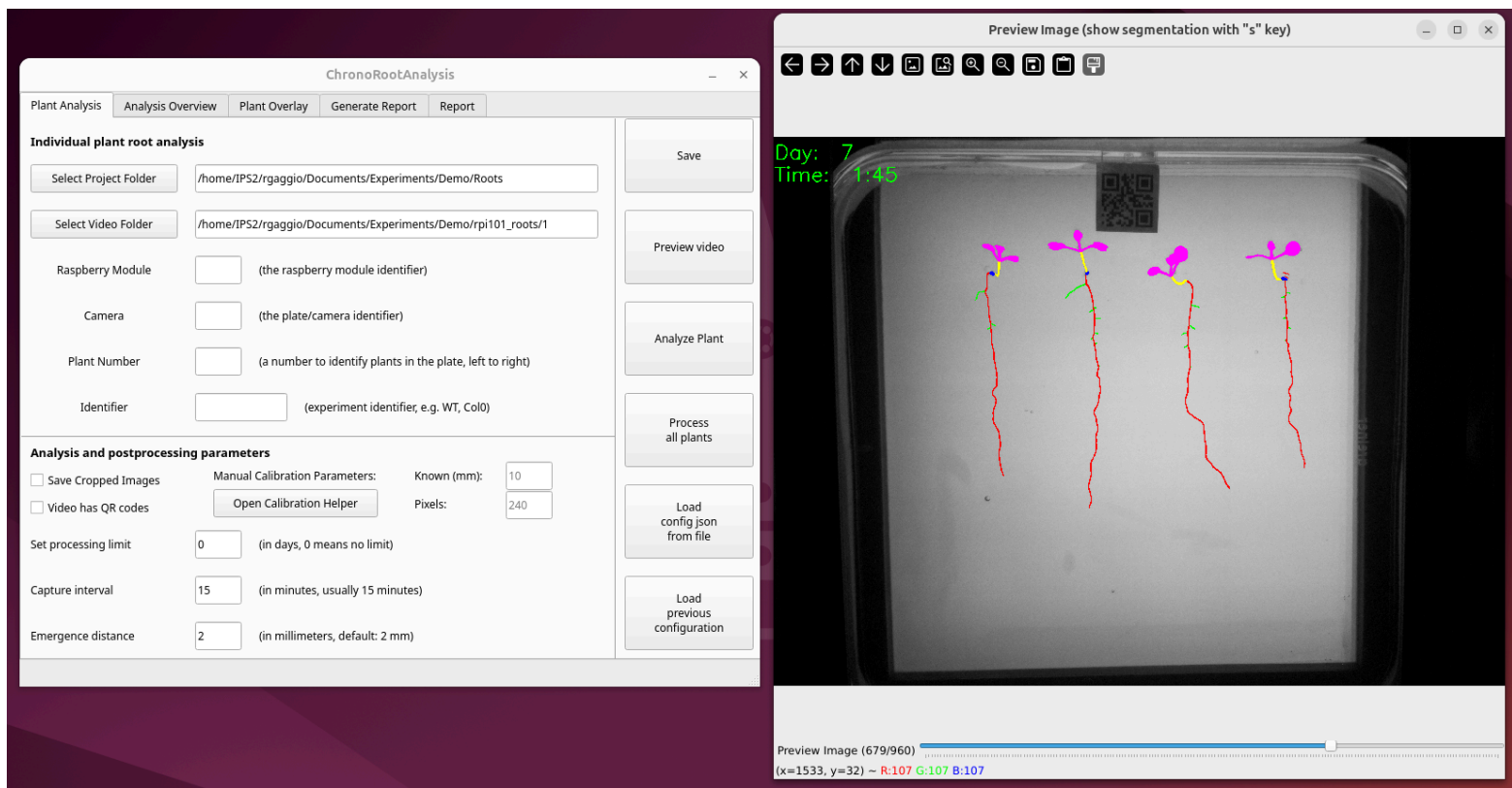
Getting Started:

Begin by creating a dedicated folder for storing results, then select the first video from the Demo dataset. A popup window will prompt you to choose the appropriate folders.



Video Preview and Setup:

Click the "Preview Video" button to view your selected video. Use the scrollbar to navigate between frames and press the "S" button to toggle the segmentation overlay for visual inspection. In this case, we can see that the video contains a QR code.



Complete the plant identification details for the first plant you wish to analyze. The plant number is usually set up to identify plants from left to right in each plate.

Individual plant root analysis

Select Project Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/Roots

Select Video Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/rpi101_roots/1

Raspberry Module

101

(the raspberry module identifier)

Camera

1

(the plate/camera identifier)

Plant Number

1

(a number to identify plants in the plate, left to right)

Identifier

A

(experiment identifier, e.g. WT, Col0)

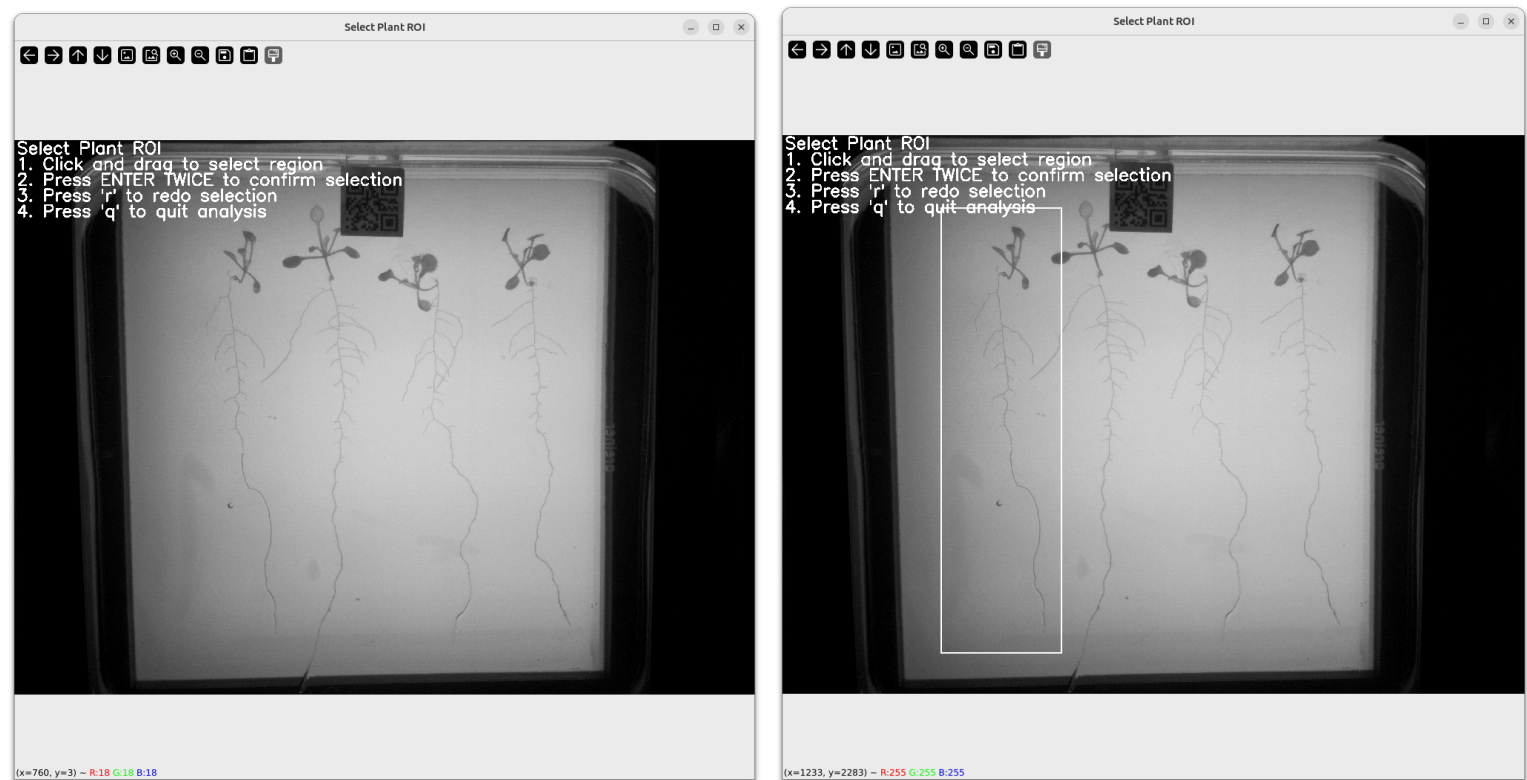
Analysis and postprocessing parameters

☐ Save Cropped Images

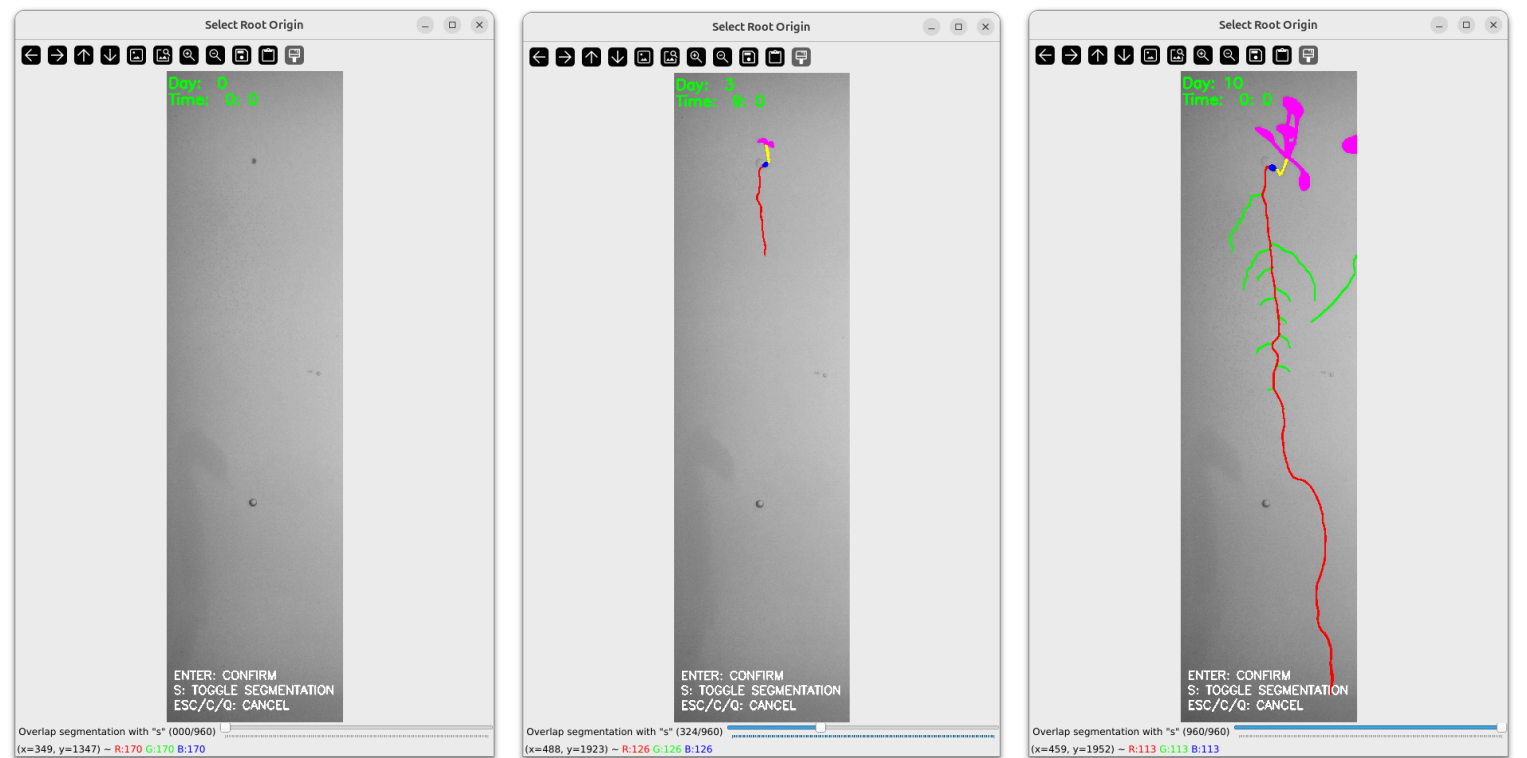
☒ Video has QR codes

Plant Analysis Workflow:

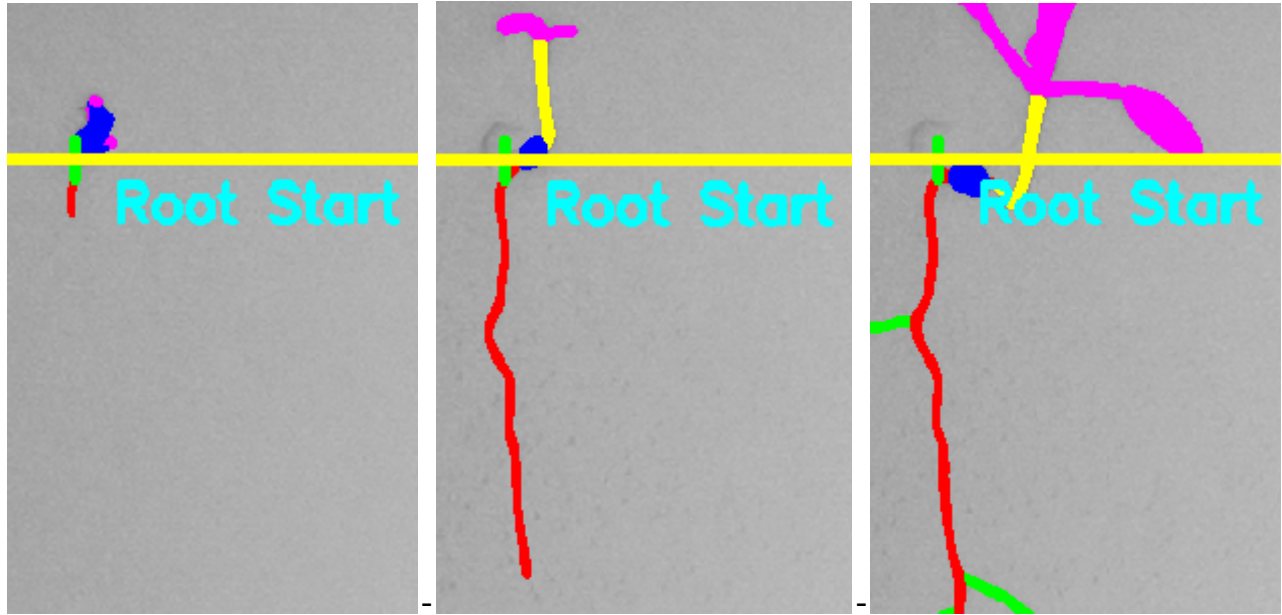
1. ROI Selection: Click "Analyze Plant" to open the ROI (Region of Interest) selector. Manually define the analysis area and press double-enter to confirm your selection.



2. Evaluate the segmentation quality: Use this step to determine the correct amount of days that the segmentation can be properly analyzed. In some cases here we can determine any potential issues like plants or leaves falling down. In this case, for example we can see that after the 10th day lateral roots will be touching the next plant.

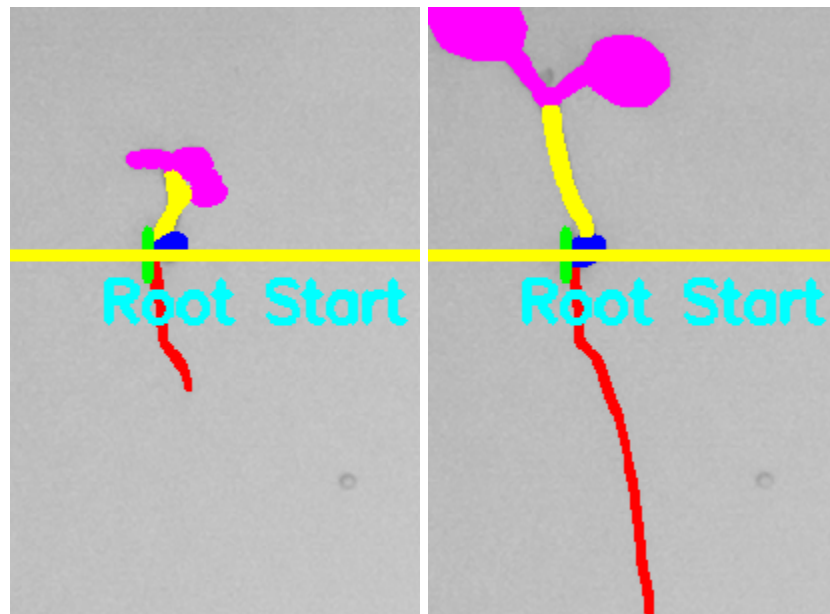


3. Select the point of main root start: This acts as a cleaning step, everything above the selected point will be discarded and not regarded as part of the root system. Please note that you can zoom in using the mouse wheel.

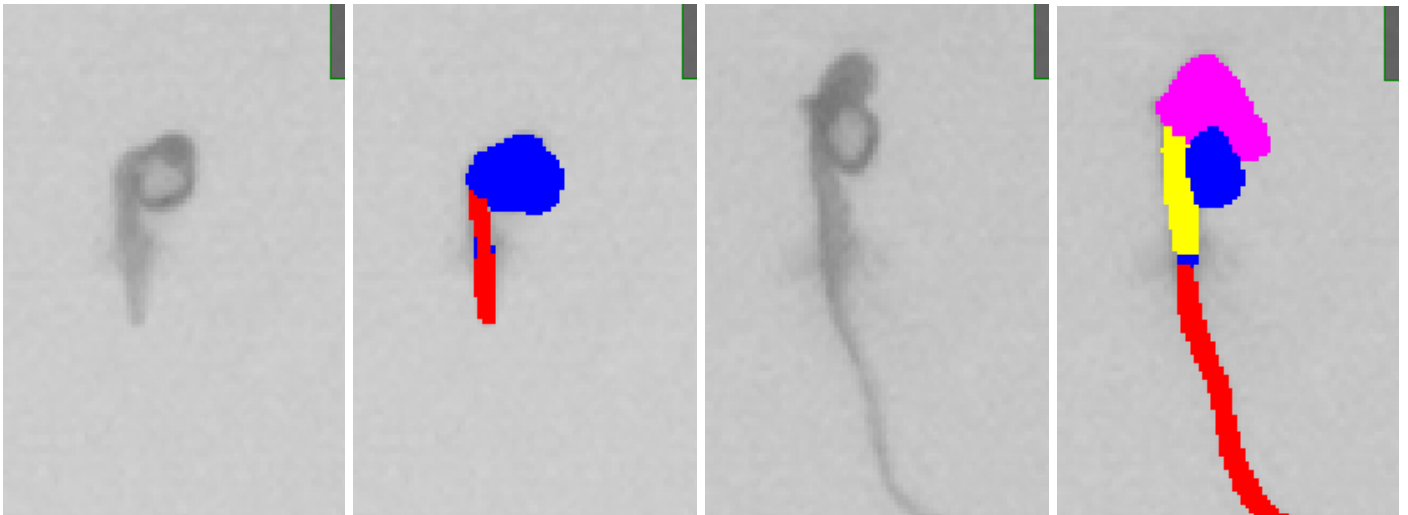


4. Repeat Process: Complete this workflow for all 4 plants in the demo. Feel free to compare varieties using patterns like A-B-A-B or A-B-C-D to observe behavioral differences.

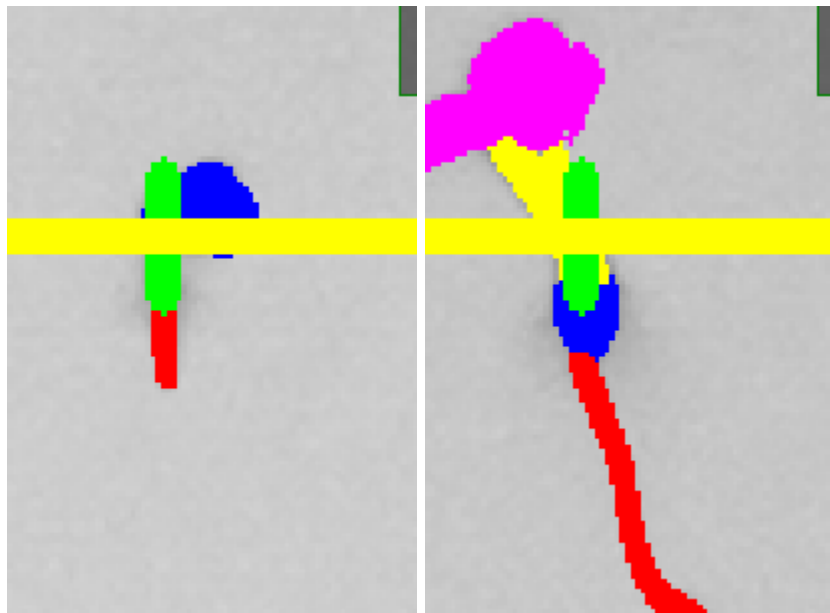
Plant 2:



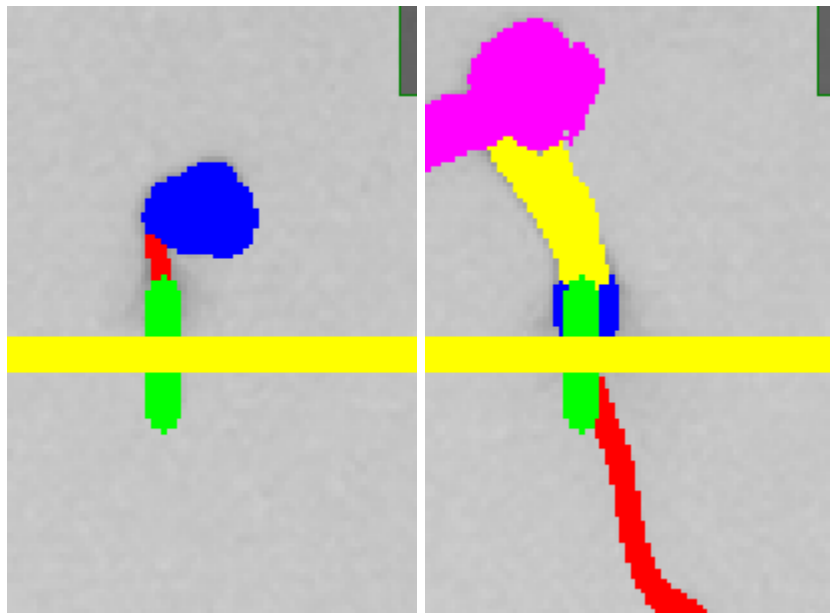
Plant 3 can help us visualize a possible issue, where the segmentation model confuses main root with hypocotyl in the early developmental stages:



A bad main root start positioning will be using the first picture or early pictures:



A good main root start positioning will take into account visual inspection to find the correct place:



Quality Control and Batch Processing:

Navigate to the "Analysis Overview" tab to monitor experiment progress and error rates. Poor ROI selection or incorrect seed point positioning may cause errors in graph creation. Once individual plant processing is complete, select "Process all plants."

PS: In case you close the app while it was analyzing, the repeat analysis button is there for you! Just remember to delete the previous plant after relaunching it.

ChronoRootAnalysis

Plant Analysis

Analysis Overview

Plant Overlay

Generate Report

Report

	Experiment	Raspberry	Camera	Plant Number	Result ID	Error Rate	Status	Finish Date
1	A	101	cam_1	plant_1	Results_0	0.002	Finished	2026-01-13 17:04:38...
2	B	101	cam_1	plant_2	Results_0	0.0	Finished	2026-01-13 17:18:05...
3	A	101	cam_1	plant_3	Results_0	0.0	Finished	2026-01-13 17:17:30...
4	B	101	cam_1	plant_4	Results_0	0.0	Finished	2026-01-13 17:04:00...

Refresh

Open Path

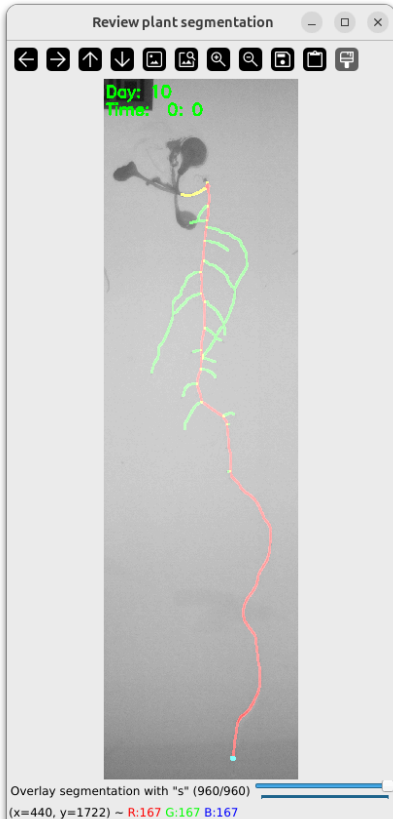
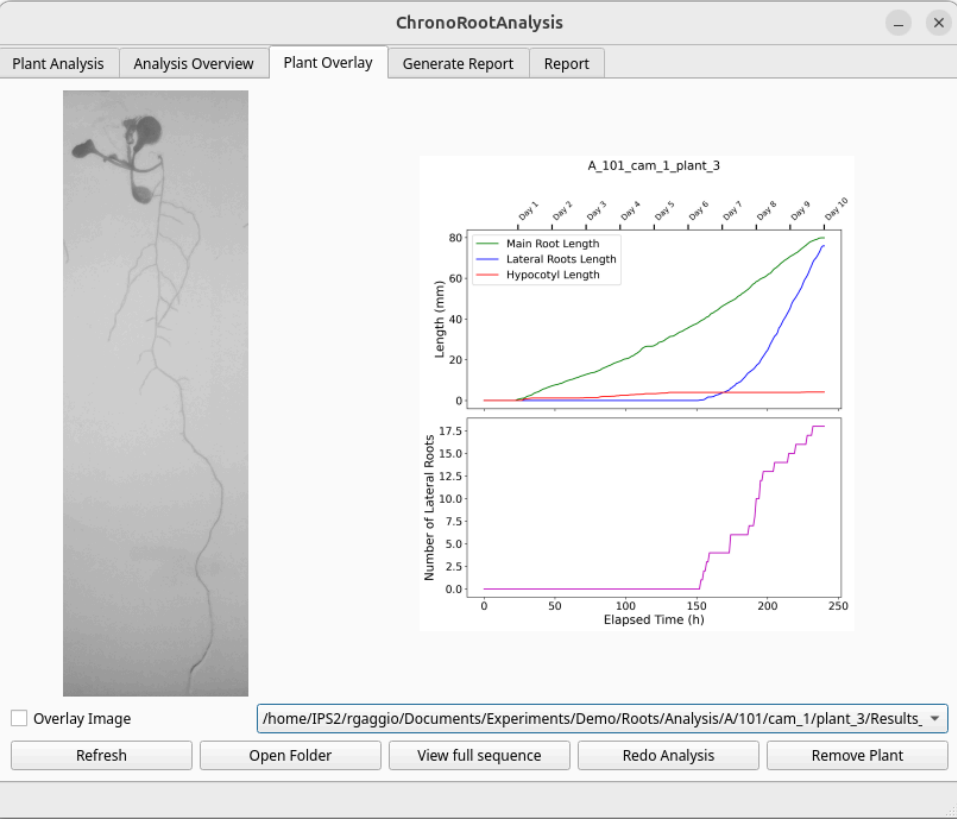
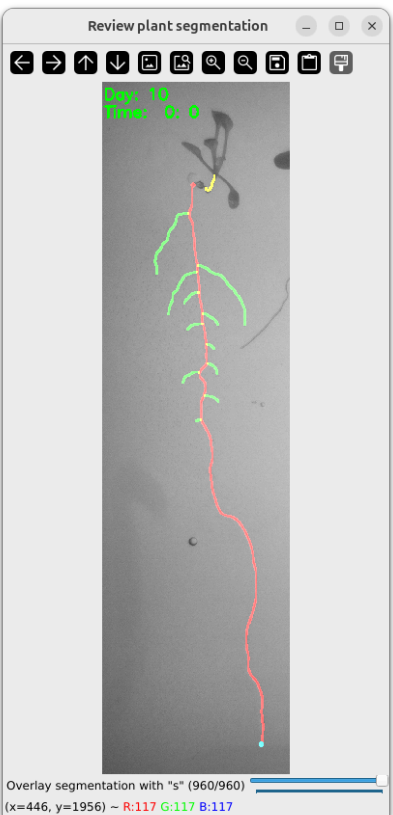
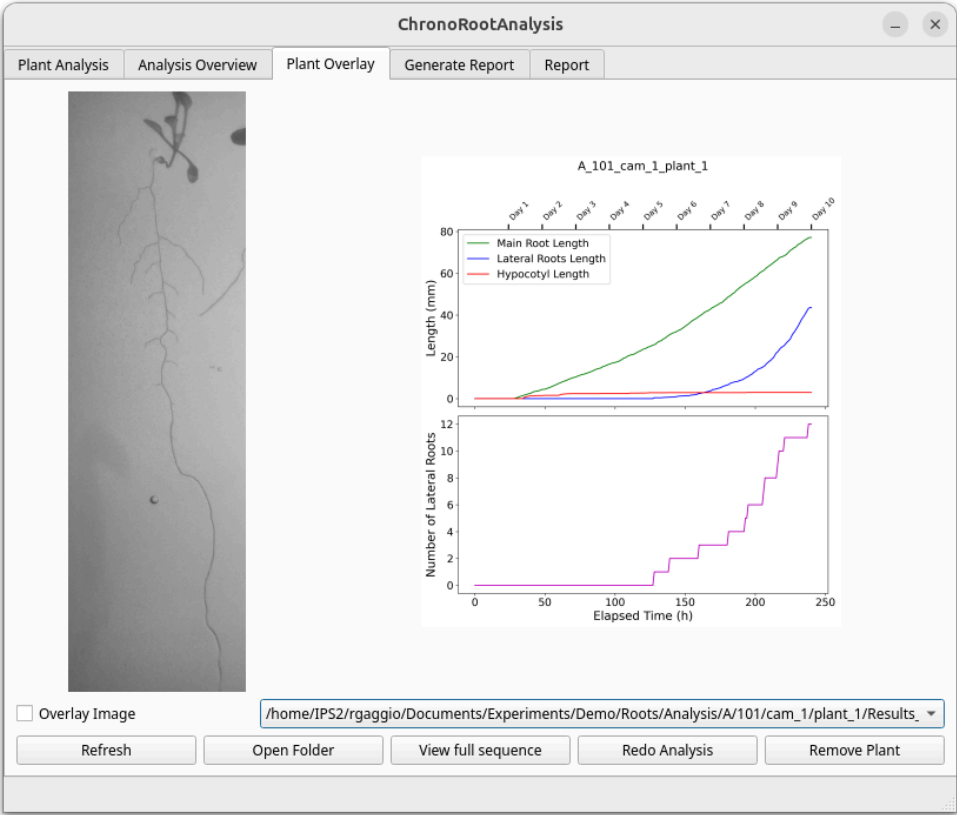
Remove Plant

Repeat Analysis

Process all plants

Visual Inspection:

The "Plant Overlay" tab provides visual inspection capabilities for both segmentation results and plant root performance analysis. This interface allows you to discard problematic plants or restart the entire process if needed. Redoing the analysis will make you choose the ROI and root start position from scratch.



Report Generation:

After completing individual plant analyses, you can generate comprehensive reports. The demo dataset includes 10 complete days of plant growth data—feel free to experiment with different report types. The system can perform interval testing (for example, every 6 hours) to compare different varieties using statistical methods such as the Mann-Whitney U test to determine if varieties show statistically significant differences.

PS: If working with longer datasets and wanting to limit processing time, set the processing limit to your desired number of analysis days.

ChronoRootAnalysis

Plant Analysis

Analysis Overview

Plant Overlay

Generate Report

Report

Select Project Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/Roots

☒ **Average intervals before testing**

Time series stats interval (dt, in hours)

6

Hypothesis testing uses Mann-Whitney test every dt interval. If selected, an average value will be used, or a step (i*dt) otherwise

☒ **Perform Functional PCA on time series**

☐ Normalize FPCA Boxplots

Number of components

2

☒ **Do Convex hull analysis**

☐ Save images for each day

Days to report

6,7,8,9,10

(Numbers separated by commas)

☒ **Evaluate Growth Speeds and perform Fourier Analysis**

Speeds stats interval (dt, in hours)

6

☒ **Do Lateral Root Angles Analysis**

Emergence distance

2

(in millimeters, default: 2 mm)

Days to report

6,7,8,9,10

First LR Tip Stats interval (dt, in hours)

6

Processing limit

0

Save

Process all plants

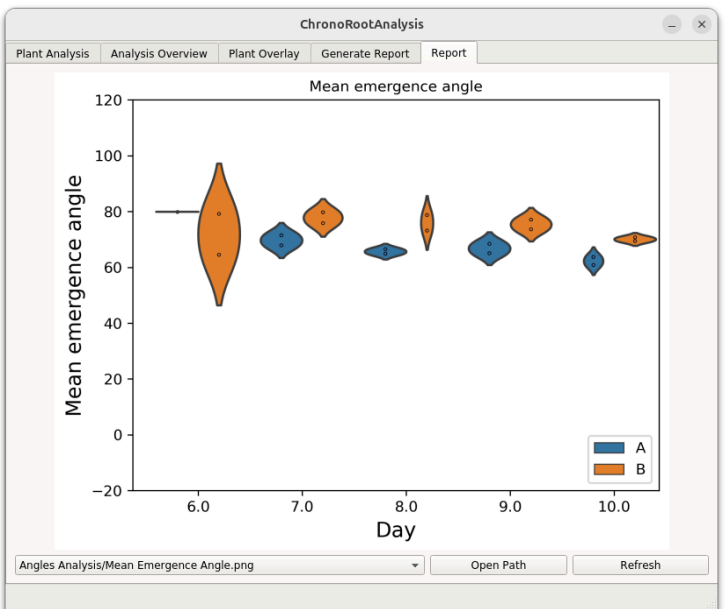
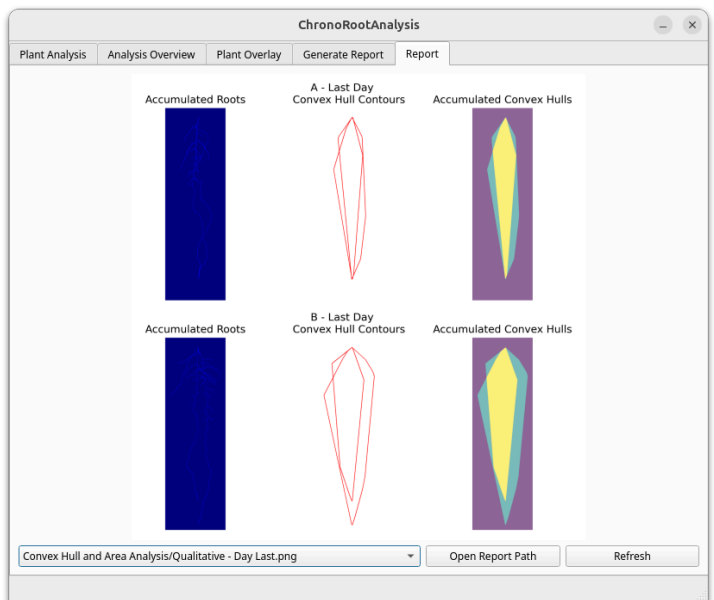
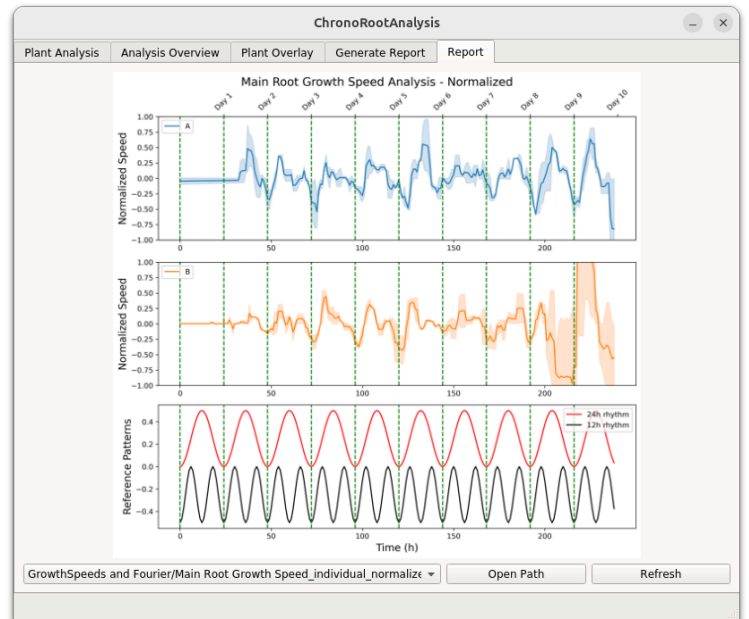
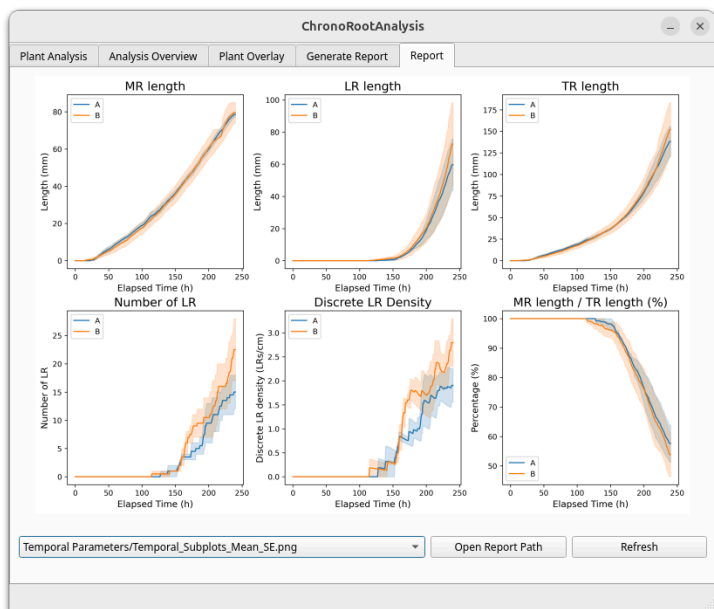
Generate report

Load previous configuration

Capture interval

15

The system automatically generates comprehensive reports containing various figure types. All raw figures and images are saved within the report directory for detailed analysis and further investigation.



< > Home / Documents / ChronoRootDATA / Demo / rpi01_roots / Results / Report



Angles Analysis



Convex Hull and Area Analysis



FPCA



GrowthSpeeds and Fourier



Individual plant plots



Temporal Parameters



Convex_Hull_Data.csv



LateralRootsData.csv



Synchronized FirstLR.csv



Temporal_Data.csv